

Predicting $\Delta\Delta G$ from Boltz-2 embeddings — generalization results

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Fast benchmark corpus: 12,359 mutations, 412 protein domains. We predict $\Delta\Delta G$ from features derived from Boltz-2 embeddings. §1 evaluates candidate feature representations and selects the **raw per-dimension Δz** ; §2 defines the corpus, model and holdouts; §3 reports the generalization results.

1. Feature representation study

1.1 What we read from the embeddings

Boltz-2 (embeddings-only) produces, for the wild-type and the mutant, three trunk tensors. At the mutated residue `i` we read three fixed objects:

- `s[i]` — the *single* representation at the site: a **384-dim** vector.
- `z[i, i, :]` — the *pair* representation of the site with itself: a **128-dim** vector.
- `z[i, :, :]` — the site's *interaction row* with all L residues: **L×128**, variable-length in L.

Everything is computed for WT and mutant and compared.

1.2 The representations we compared

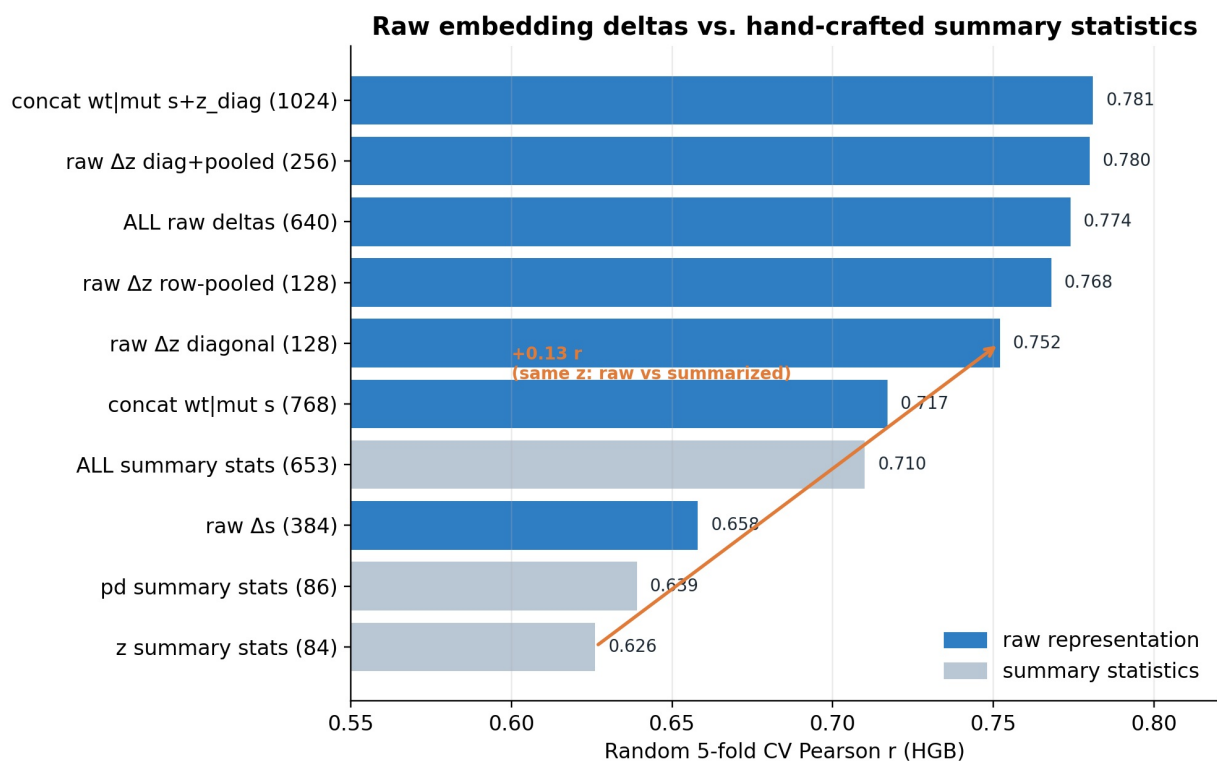
One way to build features is to reduce each slice to **hand-crafted summary statistics** (mean, SD, sum, gini, entropy, skew, ... — ~14 numbers per slice, flattened). We compared that against feeding the model the **raw values** instead:

Representation	What it is	# features
summary statistics	~14 moments per slice, flattened	653 (all slices)
raw Δs	<code>mut_s[i] - wt_s[i]</code> , all dims	384
raw Δz diagonal	<code>mut_z[i,i] - wt_z[i,i]</code> , all dims	128
raw Δz row-pooled	mean over residues of <code>$\Delta z[i, :, :]$</code>	128
raw Δz (diagonal + pooled)	the two above	256
concat WT mut	raw WT and mutant vectors, not the delta	768-1024

The `z[i, :, :]` row is variable-length (L residues), so it must be pooled to a fixed size — we mean-pool it over residues (128 dims). The diagonal and `s` are already fixed-length, so their raw deltas go in directly.

1.3 Comparison — raw beats summaries decisively

Random 5-fold CV, HistGradientBoosting, on the fast corpus:



Raw vs summary

RMSE and MAE are in kcal/mol ($\Delta\Delta G$ SD = 1.02).

Features	#	CV r	RMSE	MAE
z summary stats	84	0.625	0.793	0.585
pd summary stats	86	0.639	0.782	0.578
raw Δs	384	0.658	0.769	0.575
ALL summary statistics	653	0.710	0.716	0.526
concat WT mut s	768	0.717	0.715	0.530
raw Δz diagonal	128	0.752	0.669	0.490
raw Δz row-pooled	128	0.768	0.650	0.474
raw Δz (diagonal + pooled)	256	0.780	0.636	0.461
raw Δs + raw Δz	640	0.774	0.644	0.467
concat WT mut (s + z-diagonal)	1024	0.781	0.636	0.464

1.4 Why we use raw Δz (diagonal + pooled) for everything below

1. **Raw z recovers +0.13 r over summarized z** (0.626 → 0.752 for the diagonal alone) — *the same information*; the summary statistics were discarding most of it.
2. **Raw Δz (256 features) beats the entire 653-feature summary pipeline** (0.780 vs 0.710) — **+0.07 r with a quarter of the features.**
3. **s is redundant with raw z** — adding raw Δs on top does not help (0.780 → 0.774). So we drop s entirely; the earlier “+0.04 from s” held only against the weak summary baseline.
4. **Concatenating WT||mutant ≈ using the delta** once z is raw (0.781 vs 0.780), at 4× the features — so the explicit delta is the right, leaner choice.

Chosen representation for §2: raw Δz = [diagonal (128) || mean-pooled row (128)] = 256 features, no s, no summary statistics.

2. Setup

2.1 Data & corpus

- **Source:** Tsuboyama et al. 2023 mega-scale stability dataset, single mutants.

- **Proteins:** 412 domains, 32–72 residues — **256 natural** (PDB structures) and **156 designed / de-novo** proteins.
- **Corpus (this report):** all 412 proteins, ~30 mutations each (stratified by substitution) = **12,359 mutations**. “Wide but shallow” so the protein/de-novo holdouts see many proteins while the chemistry holdouts still cover 360+/380 substitution types.
- **Target:** $\Delta\Delta G$ in kcal/mol, **positive = destabilizing** (the mutant folds less stably). Distribution: mean 0.76, SD 1.02; ~46 % of mutations are near-neutral ($|\Delta\Delta G| < 0.5$).

2.2 Model & metric

- **Regressor:** HistGradientBoostingRegressor (median-impute → standardize → HGB), fed the 256 raw- Δz features. HGB is used rather than SVR because the holdouts require hundreds of model refits and SVR does not scale.
- **Metric: pooled out-of-fold Pearson r** — the model predicts each mutation only when that mutation is in the test split; all held-out predictions are then gathered and correlated with experiment in one pass. (Also reported: Spearman, RMSE, MAE.) “Pooled” means one correlation over all points, so it rewards getting each group’s *baseline* $\Delta\Delta G$ right, not only the within-group ranking.

2.3 The holdouts — what each one tests

Every “holdout” is just a different rule for splitting the same 12,359 mutations into train and test. They probe increasingly demanding kinds of generalization:

Holdout	What is held out of training	How the split is built	Folds	What it tests
random	individual mutations	10-fold cross-validation	10	<i>interpolation ceiling</i> — the easiest case; the model may see other mutations of the same protein
protein	whole proteins	5-fold CV grouped by protein (a protein’s mutations are entirely in train or test)	5	generalization to proteins never seen in training
cluster (homology)	clusters of homologous proteins	cluster the WT sequences by identity (greedy, e.g. 30% id → 147 clusters); 5-fold grouped so a cluster is never split across train/test	5	generalization <i>out of homology</i> — no near-duplicate protein in training
de-novo	natural vs. designed	train on the 256 natural proteins → test on the 156 designed (and the reverse)	2	transfer out of the evolutionary distribution (the hardest case)
leave-1-substitution	one substitution type A→B	hold out <i>all</i> A→B mutations, train on every other substitution	332	is a substitution’s effect inferable from other substitutions ?
source residue	all mutations <i>from</i> a residue	hold out all X→*, one source residue X at a time	20	understanding of the original residue
target residue	all mutations <i>to</i> a residue	hold out all *→X, one target residue X at a time	20	understanding of the introduced residue
chemistry class	one physicochemical class (the 13 classes are defined in §2.4)	hold out one class at a time, train on the rest	12	do physicochemical rules transfer?

The logic runs top to bottom: *random* asks whether the features carry any $\Delta\Delta G$ signal at all; *protein* and *de-novo* ask whether that signal is a real, transferable mapping or just memorized proteins; the *substitution / residue / chemistry* holdouts ask whether the model learned transferable chemistry rather than per-substitution lookups. A model that only memorized would score high on *random* and collapse on the others.

2.4 The chemistry categories

Mutations are bucketed by amino-acid chemistry, using these residue groups: **hydrophobic** = A V L I M F W Y · **polar** = S T N Q · **positive** = K R H · **negative** = D E, with the three “special” residues **G, P, C** treated on their own. Every mutation falls into **exactly one** category — they are checked in the precedence order below (proline / glycine / cysteine first, then hydrophobicity, then charge), so e.g. A→P counts as “X→P”, not “hydrophobic change”.

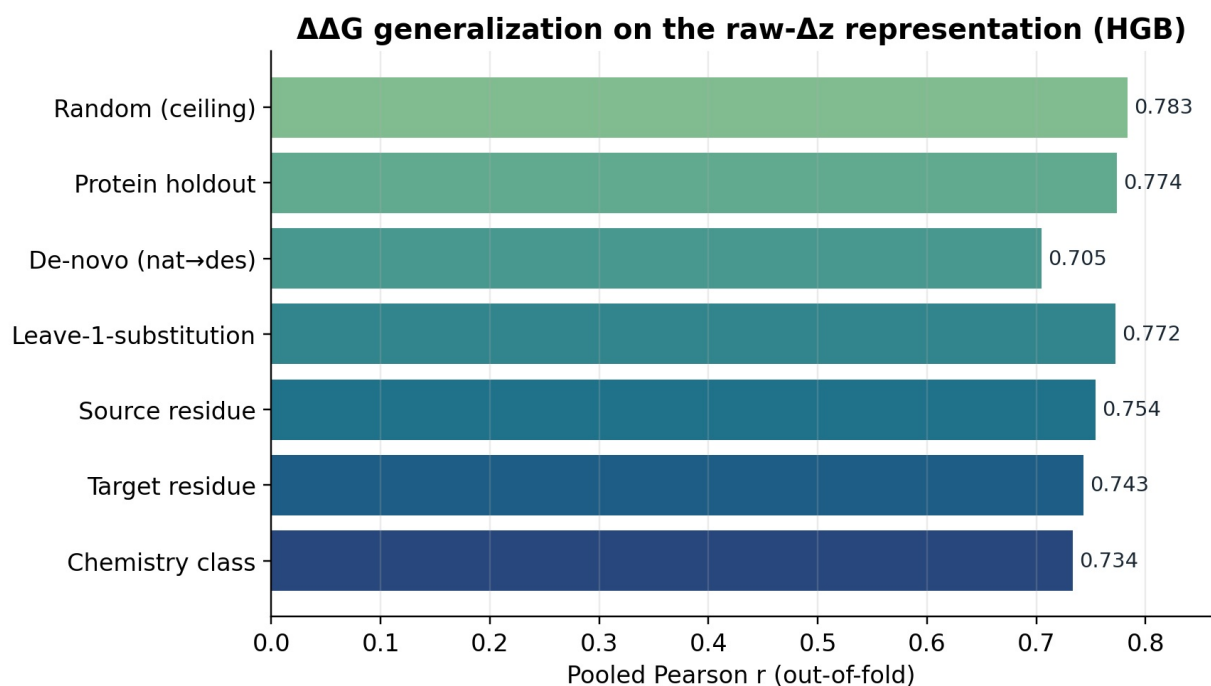
Category	Definition	n
X→P	any → proline (introduces a backbone constraint)	472
P→X	proline → any (removes it)	347
X→G	any → glycine (adds flexibility)	621
G→X	glycine → any (removes flexibility)	756
X→C	any → cysteine	489
C→X	cysteine → any	<i>dropped (only 1 example)</i>

hydrophobic→polar	A V L I M F W Y → S T N Q	949
polar→hydrophobic	S T N Q → A V L I M F W Y	1,079
positive→negative	K R H → D E (charge inversion)	240
negative→positive	D E → K R H (charge inversion)	314
neutral→charged	any non-charged → K R H / D E	1,774
charged→neutral	K R H / D E → any non-charged	2,893
other	anything not caught above (e.g. hydrophobic↔hydrophobic)	2,424

So the “chemistry class” holdout is **12 experiments** (all but C→X): each trains on 11 classes and predicts the held-out one.

3. Generalization results

Each holdout is a train/test split of the 256-feature raw- Δz table, fit with HGB.



raw- Δz holdouts

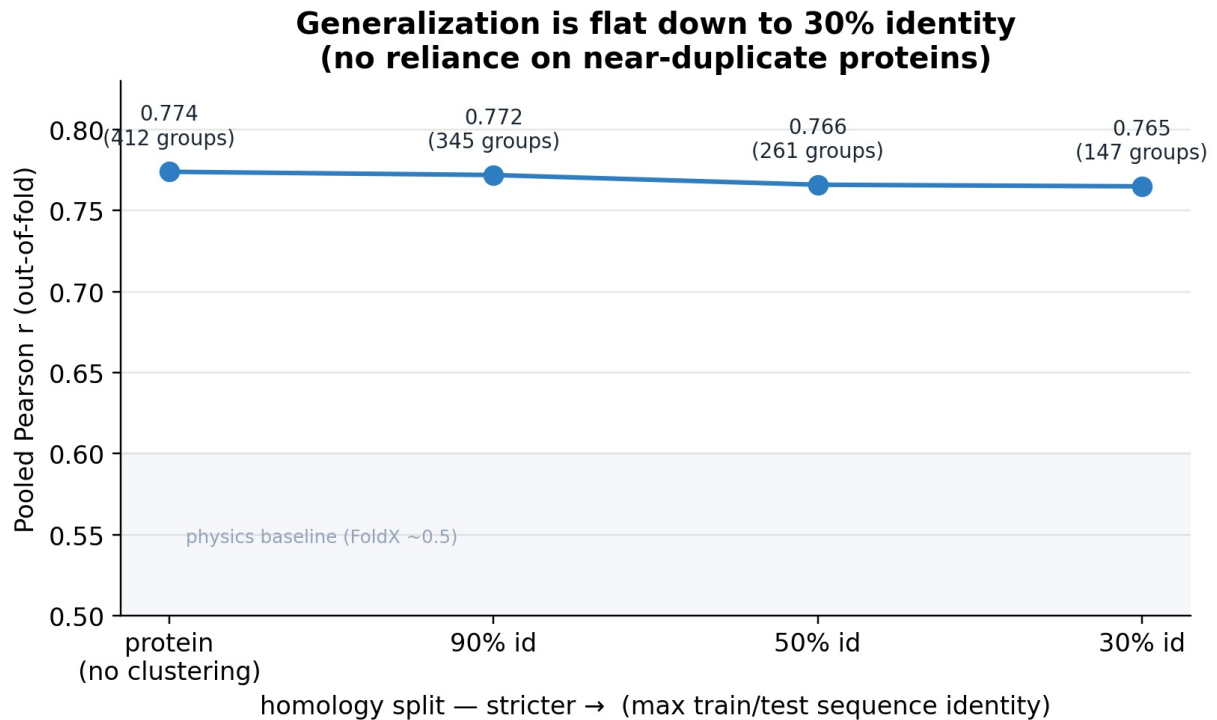
RMSE and MAE are in **kcal/mol** ($\Delta\Delta G$ SD = 1.02 kcal/mol, for scale).

Holdout	Pearson r	Spearman	RMSE	MAE	reads as
random (ceiling)	0.783	0.761	0.632	0.458	raw- Δz predicts $\Delta\Delta G$
protein	0.774	0.755	0.643	0.466	generalizes to unseen proteins
de-novo (natural→designed)	0.705	0.704	0.724	0.525	transfers to designed proteins
leave-1-substitution	0.772	0.749	0.642	0.466	transferable residue chemistry
source residue	0.754	0.728	0.667	0.488	role of the original residue
target residue	0.743	0.722	0.680	0.492	role of the introduced residue
chemistry class	0.734	0.712	0.691	0.501	physicochemical rules transfer

The random→protein gap is tiny (0.783 → 0.774) — the model generalizes to proteins it never trained on rather than memorizing them. De-novo transfer to designed proteins (0.705) is the hardest case and still strong; every chemistry and substitution holdout stays ≥ 0.73 .

3.1 Homology holdout — generalization vs. sequence identity

The protein holdout stops a protein's *own* mutations leaking between train and test, but Tsuboyama also contains **near-duplicate** entries (e.g. ubiquitin scanned with different background mutations, ~99 % identical). To rule out that the model rides on these, we cluster the WT sequences by identity (greedy, CD-HIT-style) and re-run the grouped CV so **no cluster is split across train/test** — progressively stricter down to 30 % identity.

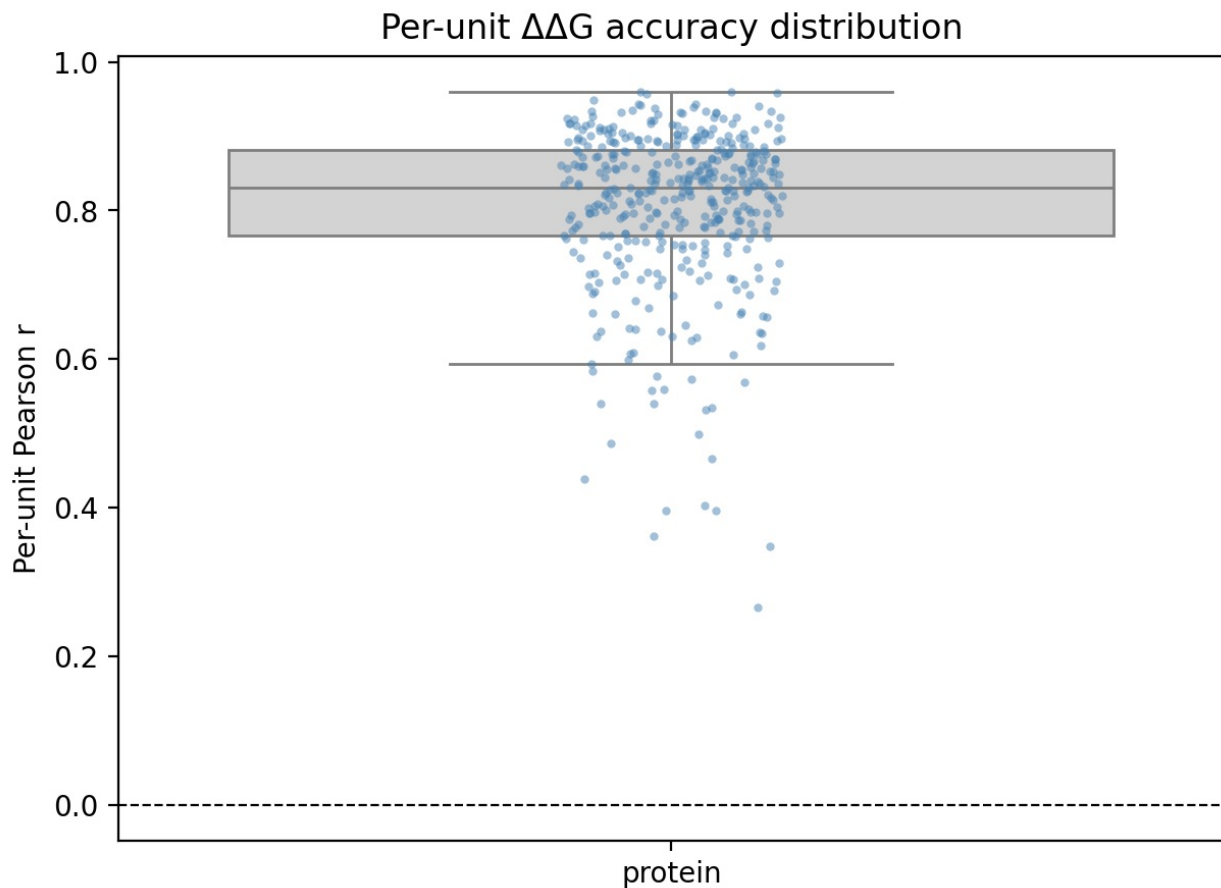


homology sweep

Split	# groups	pooled r	Spearman	RMSE	MAE
protein (no clustering)	412	0.774	0.755	0.643	0.466
cluster @ 90 % id	345	0.772	0.753	0.646	0.469
cluster @ 50 % id	261	0.766	0.748	0.653	0.473
cluster @ 30 % id	147	0.765	0.747	0.655	0.475

Accuracy is essentially flat — a strict 30 %-identity split costs only -0.009 vs. the protein holdout, and 91 % of the 147 clusters still score $r > 0.7$. The model is not exploiting near-duplicate proteins; it generalizes to sequences with no close homolog in training.

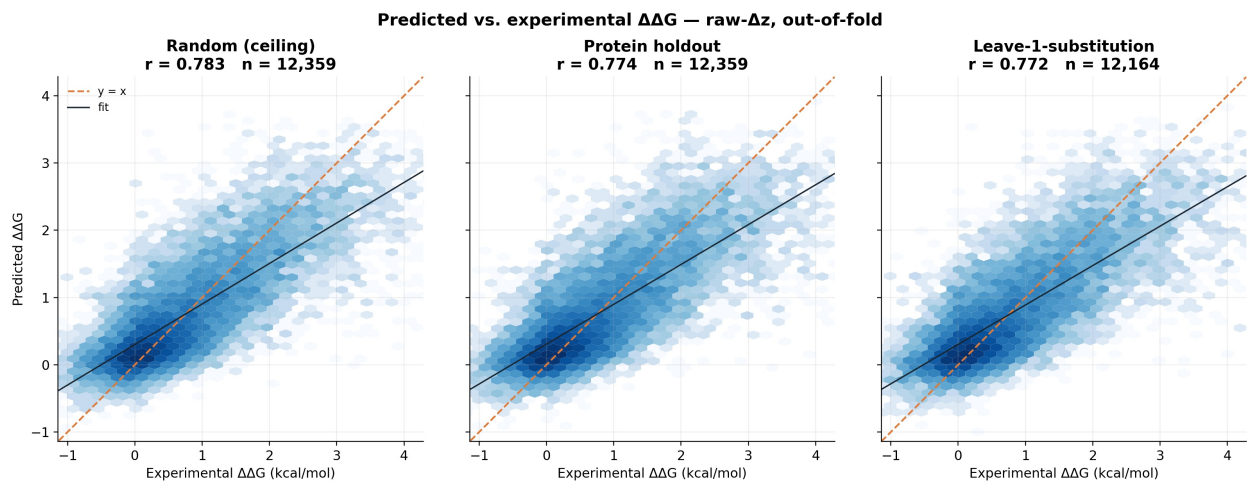
3.2 Per-protein distribution



per-protein

Across the 412 held-out proteins: per-protein r **mean 0.806, median 0.831, SD 0.109; 87 % score $r > 0.7$, none below 0.3, none below 0.** No family where the model silently fails.

3.3 Predicted vs. experimental



regression

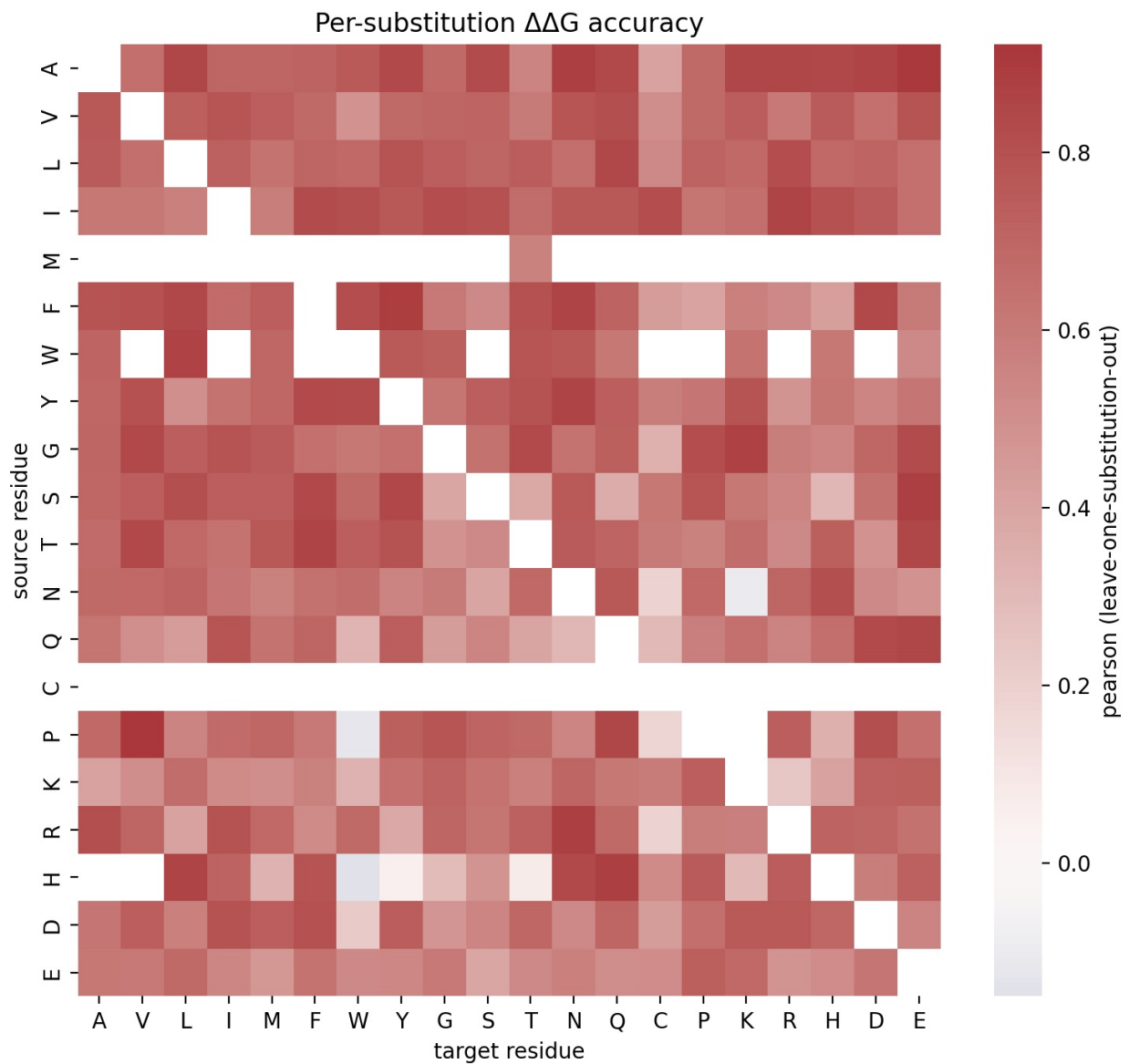
Out-of-fold predictions for the random, protein and substitution holdouts — nearly identical, the visual form of the generalization result. (The fit line is still slightly shallower than the diagonal: some regression-to-the-mean on the most destabilizing mutations remains.)

3.4 Chemistry transfer

Chemistry is probed at **three levels of granularity** — individual substitutions, single residues, and broad physicochemical classes.

(1) Leave-one-substitution-out — each of the 332 individual A→B substitution types held out in turn — gives $r = 0.772$ pooled. The 20×20 map shows the per-cell accuracy (source residue $\times$ target residue); note that holding out one *cell* still lets

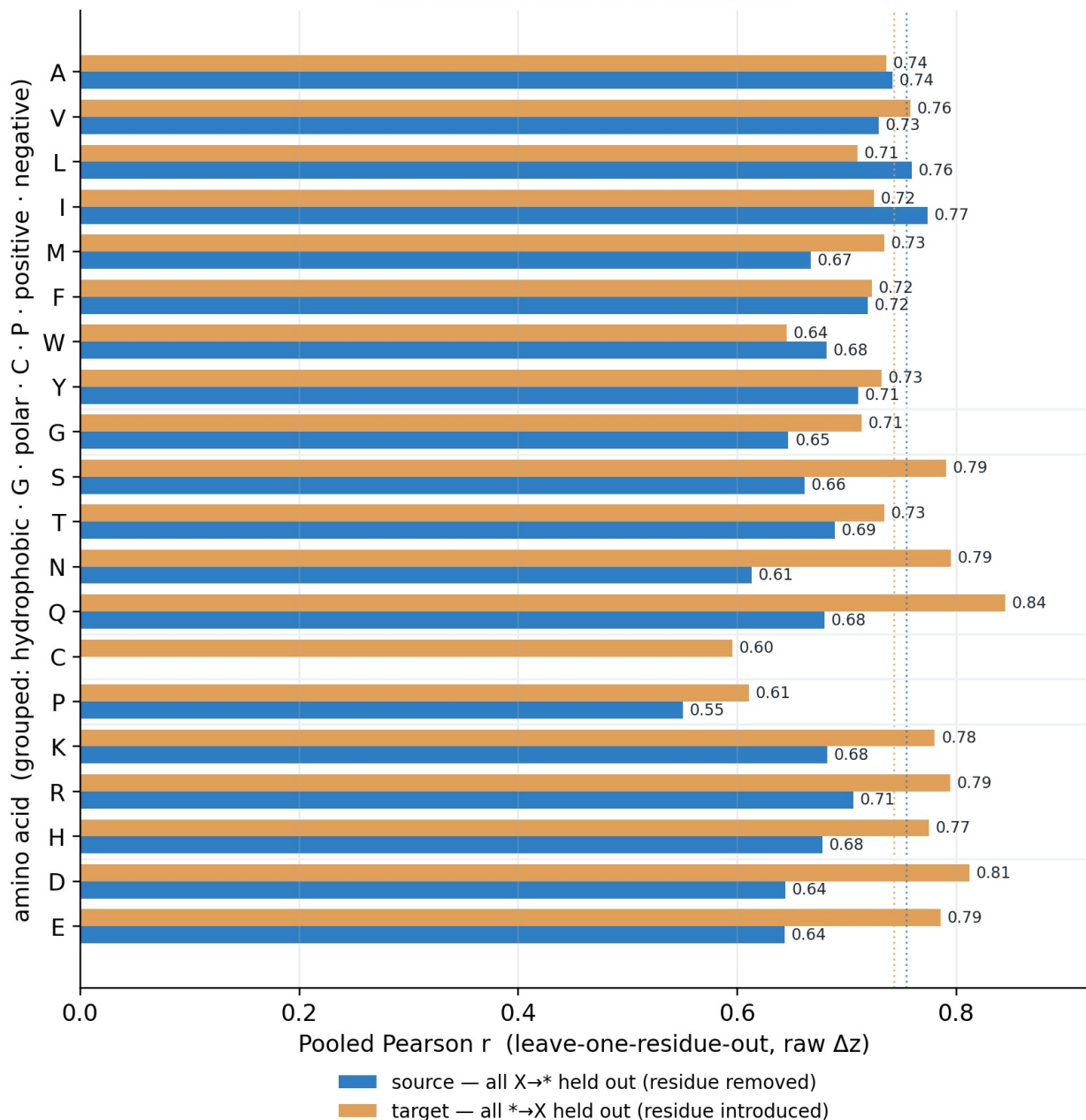
the model see both residues in other contexts:



substitution heatmap

(2) Leave-one-residue-out is stricter: hold out every mutation **from** a residue (source, $X \rightarrow *$) or **to** it (target, $* \rightarrow X$) — so the model never trains on that residue at all. This is the row/column-level holdout the substitution map does *not* capture. Pooled $r = 0.754$ (source) / 0.743 (target); per residue:

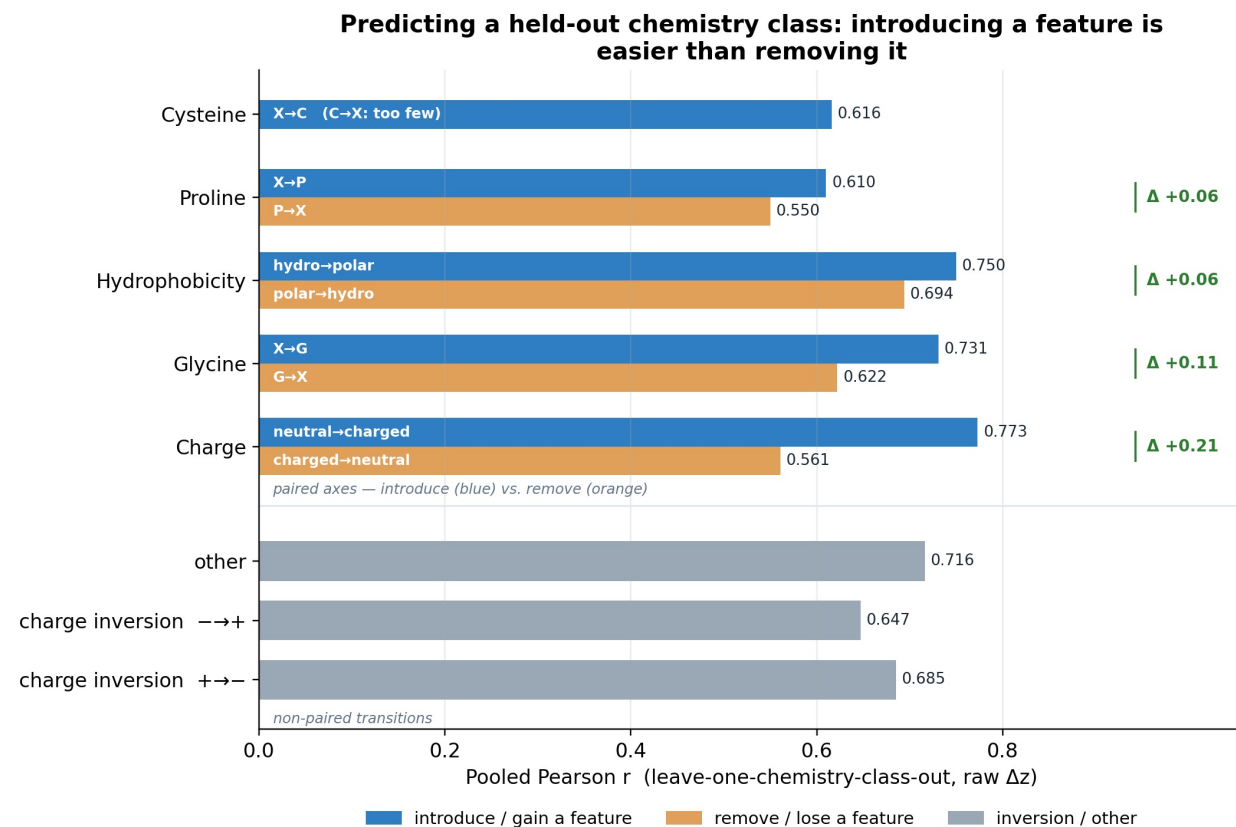
Per-residue holdouts: predicting mutations of a residue the model never trained on



per-residue holdouts

Proline is the hardest in both directions (source 0.55) — consistent with its unique backbone role — and cysteine has too few C→* mutations to score as a source. For the polar and charged residues, *introducing* a residue (target) tends to be predicted better than *removing* it (source) — the same introduce-vs-remove asymmetry seen below across chemistry classes.

(3) Leave-one-chemistry-class-out (the 12 classes of §2.4) gives $r = 0.734$ pooled. Every class — even the hardest — stays ≥ 0.55 , so the model predicts a whole physicochemical class of mutation it was never trained on. But the per-class result also reveals a consistent, interpretable pattern:



chemistry classes

Introducing a chemical feature is predicted markedly better than removing it. Across every paired axis the “introduce” direction wins — dramatically for charge (neutral→charged 0.77 vs charged→neutral 0.56, Δ 0.21), and consistently for glycine (+0.11), proline and hydrophobicity (+0.06). A plausible reading: *adding* a disruptive residue (a charge, a proline kink, a glycine hinge) into a resolved wild-type structure produces a destabilization the embedding can read off locally, whereas *removing* one depends on the specific role it played — a more context-dependent, harder-to-predict quantity. Worth flagging as an observation to probe, not yet a firm claim.

4. Conclusions

1. **The raw representation is the model to use** — raw Δz (diagonal + pooled row, 256 features), no s , no summary statistics (see §1 for the justification).
2. **It generalizes on every axis** — protein 0.774 sits just below the 0.783 random ceiling, de-novo transfer reaches 0.705, and every substitution/chemistry holdout stays ≥ 0.73 . Per-protein accuracy is tight (mean 0.806, none failing).
3. **Next:** rebuild the feature extractor to emit raw Δz directly (replacing the summary layer) and re-run the wide corpus on it; then the outstanding cluster/homology holdout.